Hake qPCR and metabarcoding data for the 2019 eDNA samples: what we have and open questions

12 October 2026

Why we are asking

The joint eDNA + visual model needs, for hake, the **raw qPCR Ct values** and the **standard curve** (intercept, slope, amplification efficiency, Ct noise) that go with them. Until now we only had copies/µL from `hake_qPCR_MURI_df.csv`, so the real-data fits used placeholder qPCR parameters. We now have `qPCR_2019A.rds` (Ole) and the Stan code from the qPCR vignette. This note summarises how those line up with the metabarcoding data already in hand, and lists the decisions we need your input on (see *Questions* below).

What we have

File	Content
`hake_qPCR_MURI_df.csv`	2,663 hake qPCR wells, 562 sample IDs, 551 bottles; copies/µL, dilution, inhibition rate, plate. **No Ct, no well ID.**
`MV1_MURI_df.csv`	1,709 metabarcoding (MV1) PCR replicates from the same 551 bottles; reads per taxon, `Rep`, template `Dilution`.
`qPCR_2019A.rds`	2019 survey (2 Jul to 17 Aug), "analysis A": field wells with **Ct**, copies/µL, dilution, inhibition rate, plate, well, `drop.sample`, filtered volume; hake **standards** on all 33 plates; field negatives; NTCs.
`qPCR/*.stan`, `qPCR_vignette_part_1.qmd`	Standard-curve models from the vignette (single- and multi-plate), field-sample model.

`qPCR_2019A.rds` is the same data as the earlier `qPCR_2019.rds` except that sample IDs now carry the project prefix (`52193_282`), bathymetry columns were revised, and `qpcr_analysis` / `project` columns were added.

How qPCR and metabarcoding correspond, bottle by bottle

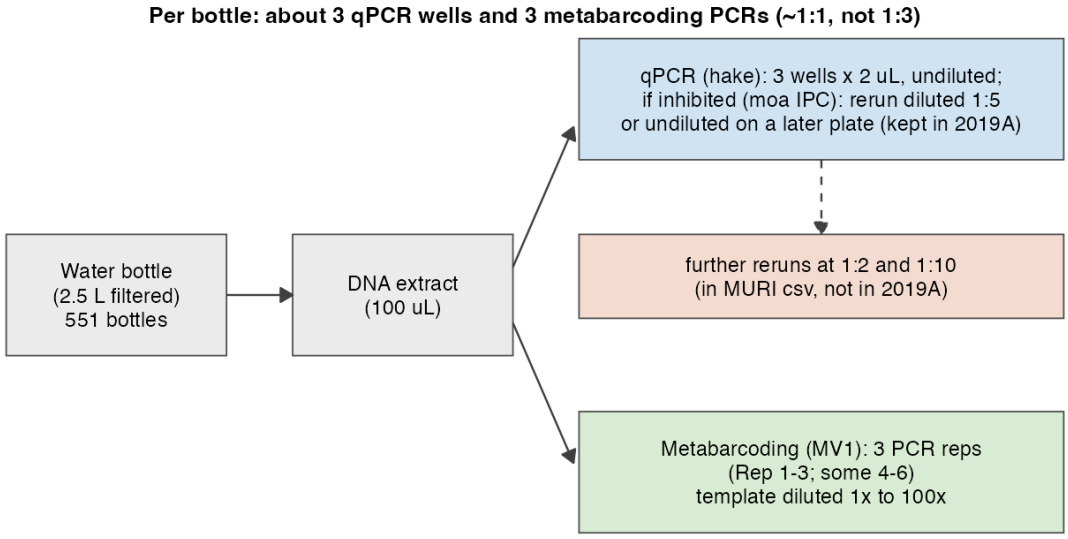


Figure 1: What happens to one water sample (counts from the files in hand).

Per bottle	qPCR (hake)	Metabarcoding (MV1)
Typical	**3 wells** on one plate, one dilution (855 of 922 sample × plate × dilution groups)	**3 PCR replicates** (417 of 551 bottles; 82 have 4)
Extras	Reruns of inhibited samples on later plates and/or at higher dilution	`Rep` 4 to 6 on some bottles
Dilution	1, 0.2 (1:5), 0.5 (1:2), 0.1 (1:10); coded as a *fraction*	1x to 100x; coded as a *fold*
Total	2,663 wells	1,709 PCR replicates

We had understood the design as **one qPCR to three metabarcoding reactions per bottle**. The files instead show roughly **3 qPCR wells to 3 metabarcoding PCRs** (about 1:1). Our reading is "one qPCR assay of the extract, run as a technical triplicate", alongside three independent metabarcoding PCRs. The overall 2,663 / 1,709 = 1.56 ratio comes from qPCR reruns, not the basic design. Eleven bottles have two qPCR sample IDs (field duplicates, or the 30EtOH pairs?).

Which qPCR wells are in analysis A

All 562 of our sample IDs are in `qPCR_2019A.rds`, and 1,673 of our 2,663 wells match it exactly (same sample, plate and inhibition rate; copies/µL agree for 99.8%). The other **990 wells are not in analysis A**. They are not random; they are all inhibition-related (Figure 2):

Wells in our csv	in 2019A	not in 2019A	Inhibition rate (median)
Undiluted (1)	1,284	217	0.69 for the 217 dropped wells (41% > 0.75)
1:5 (0.2)	389	9	0.06 for the kept wells
1:2 (0.5)	0	365	0.12 for the 0.5 and 0.1 reruns combined
1:10 (0.1)	0	399	
Total	**1,673**	**990**	

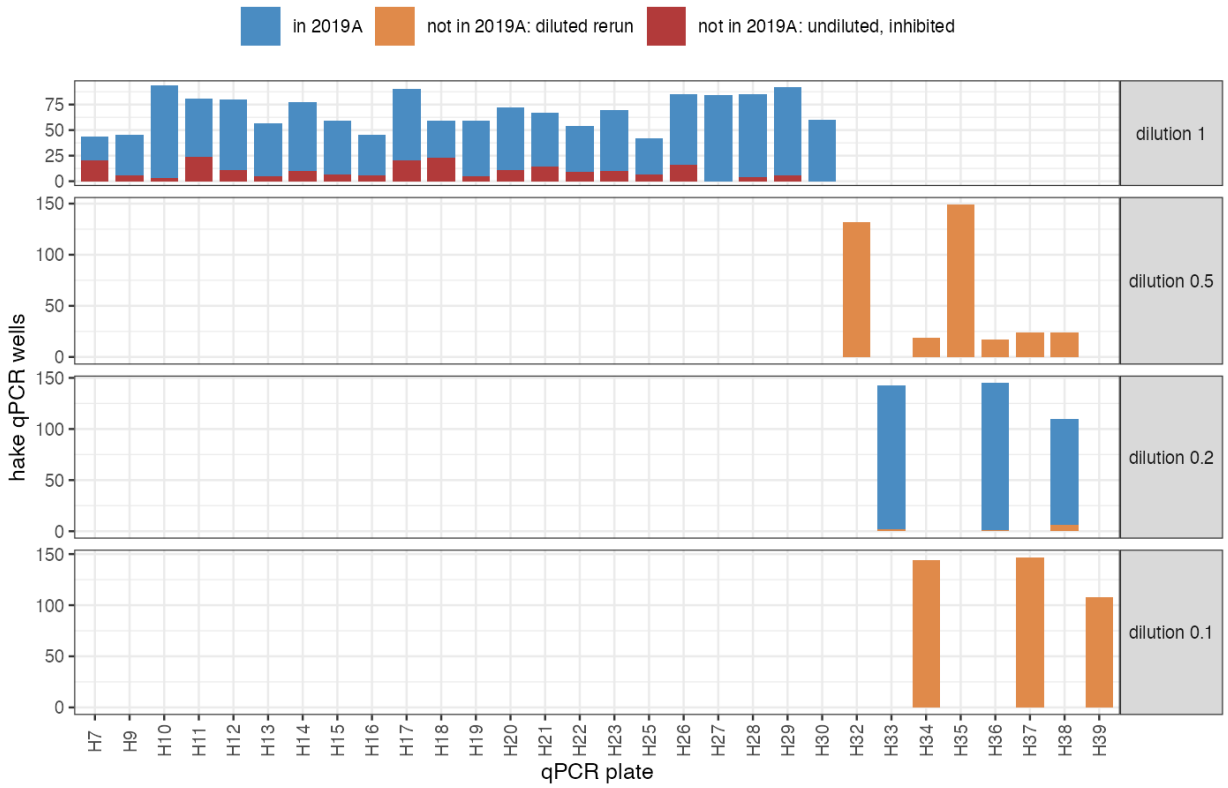


Figure 2: Hake qPCR wells in our csv, by plate and template dilution, coloured by whether they are in analysis A.

- **134 samples** were inhibited when undiluted (by the moa internal control) and rerun diluted. Analysis A keeps their **1:5 triplicate** and drops the undiluted wells and the 1:2 / 1:10 runs (plates H32, H34, H35, H37, H39 entirely, and part of H36 and H38).
- **45 further samples** had inhibited undiluted wells dropped without a diluted rerun. Analysis A keeps their other wells, mostly an undiluted rerun on a later plate (e.g. H9 to H17).
- Every bottle that was rerun for qPCR was also diluted for metabarcoding (134 of 134); 148 more were diluted for metabarcoding only. So the dilutions appear to be driven by inhibition of the extract, applied to both assays, with metabarcoding diluted more often.
- After this cleaning, almost every sample has **exactly one triplicate** in analysis A (508 of 562 samples have 3 wells); no sample is lost.

Two other things in analysis A for our samples:

- `drop.sample = "30EtOH"` on 166 wells (56 samples). We currently keep these.
- 180 of 1,426 detected wells have **Ct > 37.5** (max 44.7). The vignette source has a "Ct > 37.5 = non-detection" recode, but it is commented out.

Standard curve and Stan code

- Hake standards (gBlocks, 1 to 10^5 copies/µL) are on all 33 plates, including every plate our wells were run on: 765 wells, 106 inhibited (> 0.75, mostly 10^5), 26 non-detects.
- `qPCR_standards_multi-plate.stan` (per-plate β_0, β_1; shared $\gamma_0, \gamma_1, \sigma_0, \phi$; slope bound $\beta_1 \leq -1/\ln 2$) compiles and is what we would fit first. Its parameters map onto our qPCR hurdle as $\alpha_{ct} = \beta_0$, $\beta_{ct} = -\beta_1$, $\kappa = \phi$, with $\gamma_0, \gamma_1, \sigma_0$ unchanged (our λ = copies in the 2 µL reaction, i.e. $2K$).
- `qPCR_simple.stan` (standards + field samples jointly) does not compile as sent (old array syntax, `bin_obs` vs `bin_obs_stand`); the fixes are small. `qPCR_controls.stan` and `qPCR_standards_multi-plate_nobin.stan` look like work in progress.
- None of the compiling models has a **dilution offset**. We would add log(dilution) to $\log \lambda$ for each well, so a 1:5 well is not read as a sample with 5x less DNA.

Questions

1. **Design.** Is it right that each bottle got one qPCR assay (technical triplicate) and three metabarcoding PCRs? What are the eleven bottles with two qPCR sample IDs?
2. **Reruns.** Is analysis A deliberately "one clean triplicate per sample"? Should we drop the undiluted inhibited wells and the 1:2 / 1:10 reruns too, or are the 1:2 / 1:10 runs usable extra information (with a dilution offset)?
3. **30EtOH.** Should samples flagged `30EtOH` / `30EtOHpaired` be excluded?
4. **Late Ct.** Should detected wells with Ct > 37.5 count as detections or as non-detections?
5. **Inhibition cutoff.** Is 0.75 the threshold for field samples as well as standards, and is analysis A already filtered on it?
6. **Metabarcoding dilution.** Was MV1 template dilution (5x to 100x) also inhibition-driven? Is there a reason not to treat it as a known offset on amplifiable copies?
7. **Bathymetry.** Which depth column (`bathy.bottom.depth` or `bottom.depth.consensus`) do you recommend? We currently take bottom depth from ETOPO 2022.

Answers (Ole, October 2026)

1. **Design.** Match qPCR and metabarcoding on sample number.
2. **Reruns / dilution.** Qualify dilution as an offset in the qPCR model (0.1 means 1:10). Metabarcoding needs an offset too.
3. **30EtOH.** A lab mix-up lost about 20 to 30% of the hake DNA in these samples. Either estimate the loss with hake qPCR and assume it is the same for the other taxa, or exclude them; which depends on how many were metabarcoded (56 of 569, 10%).
4. **Late Ct.** The 2019 outliers have already been dealt with; use the observations as provided.
5. **Inhibition.** The 2019 qPCR is cleaned: no inhibited observations remain, and there is one dilution per sample.
6. **Metabarcoding dilution.** An offset.
7. **Bathymetry.** Use `bottom.depth.consensus`.

Since then we have also received `M3_compiled_taxon_table_wide.csv` (reads per taxon for six markers; we use MarVer1) and `Hake_2019_metadata.csv`, and use `depth_cat` (target depth) in place of recorded depth.

What we did

- Hake qPCR: all analysis-A wells, Ct and dilution as provided (1,532 wells, 513 bottles after dropping the 30EtOH samples for now).
- Standard curves per plate, estimated from the hake standards (`stan/qpcr_standards.stan`, after the multi-plate vignette model): intercepts 36.1 to 39.9, slopes 1.45 to 1.49 Ct per log copy (efficiency 0.95 to 1.00), detection rate 0.90 per copy.
- Dilutions as offsets: log fraction for qPCR, minus log fold for metabarcoding. Where a bottle was metabarcoded at several dilutions, the dilution with the most reads is kept.

Follow-up questions

- **Hake in field negatives.** 73 of 114 evaporated-DI and 25 of 72 lab-DI field-negative wells detect hake (up to about 12 copies/µL); extraction and PCR controls are clean. Is this known, and should field samples be corrected for it?
- **Metabarcoding template volume.** How many µL of extract went into each MarVer1 PCR? *Answered (2026-10-22): always 2 µL, as for qPCR.*
- **Minimum read depth.** We treat a MarVer1 replicate with no reads at all as a failed PCR and drop it (in 7 bottles every replicate failed). Of the replicates we keep, 136 have fewer than 1,000 reads in total and 47 fewer than 100; they detect hake less often (49% under 100 reads, 69% over 10,000). Do you use a minimum total-read cutoff below which a replicate counts as failed, and if so, what is it? *Answered (Pedro, 2026-10-22): 1,000 reads summed over a sample's replicates. At the best dilution this drops 6 more bottles' metabarcoding (each with one replicate with reads, 1 to 540 reads); their qPCR is kept.*

Numbers and figures: `docs/dataQs/make_figs.R`.